

RECEIVER-INDUCED MOMENT EXCHANGE FOR DISTRIBUTED PROJECTED KLA-LMB TRACKING

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ABSTRACT

In the studied distributed tracker, the network interface transmits full Gaussian-mixture (GM) messages even though the receiver discards mixture structure before fusion. We treat this mismatch as a message-interface problem for a projected Kullback–Leibler-average labeled multi-Bernoulli (KLA-LMB) receiver. Factoring its implementation into a canonical label-wise moment projection and downstream fusion identifies a receiver-induced equivalence class. Under matched active labels, Metropolis weights, delivery masks, and numerical conventions, sender-side projection preserves every fused existence probability, mean, and covariance. We realize the interface with a versioned binary application-layer codec. In 50 paired eight-sensor trials, moment exchange reduced attempted payload bytes by 58.28% (95% bootstrap interval: 57.92–58.64%). All 1,119,037 audited label-instance outputs were identical. The certificate removes representation-level traffic without changing this receiver’s output; it does not extend to mixture-aware or multi-round consumers.

Index Terms— Distributed multi-object tracking, labeled random finite sets, LMB filter, KLA fusion, receiver-induced message

1. INTRODUCTION

Distributed multi-object tracking combines complementary sensor views without centralizing measurements. Labeled multi-Bernoulli (LMB) filters jointly represent target existence, state uncertainty, and identity [1, 2, 3]. A Gaussian-mixture (GM) LMB message may contain several components per label. The studied implementation serializes every component although its receiver immediately collapses each label to one Gaussian. It therefore transmits fields this receiver cannot observe.

Communication-aware RFS trackers reduce transmission frequency, select mixture components, or exchange projected state estimates [4, 5, 6, 7, 8]. These methods decide when to send, which components to share, or which compact estimate

to form. We instead ask a consumer-facing question: for a fixed receiver, which transmitted fields can affect its output?

We study a projected Gaussian KLA-LMB receiver with one synchronous neighbor- fusion round per filtering step. It aligns labels, normalizes Metropolis weights over sources carrying each label, projects each GM Bernoulli to existence and two moments, and fuses the resulting Gaussians. Let $\mathcal{T}$ denote the canonical encode/decode content map. For the executed receiver $\mathcal{F}_{\omega, \mathcal{R}} = \mathcal{G}_{\omega, \mathcal{R}} \circ \mathcal{P}$, the wire-path identity is $\mathcal{F}_{\omega, \mathcal{R}}(\mathcal{T}\pi) = \mathcal{F}_{\omega, \mathcal{R}}(\mathcal{T}\mathcal{P}\pi)$. The nontrivial check is that label presence, Gaussian overlap, and deterministic regularization also factor through $\mathcal{P}$ after transport (Fig. 1).

Our contributions are threefold. First, we extract an executable receiver contract that includes label presence, KLA weights, covariance canonicalization, and solve regularization. Second, a versioned application-layer codec and adversarial property tests validate the contract across serialization. Third, a frozen 50-trial paired evaluation shows a 58.28% mean attempted-byte reduction and zero residual across 1,119,037 label-instance comparisons. We claim neither equality of the underlying mixture densities nor sufficiency for a mixture-aware or multi-round receiver.

2. RELATED WORK

Random finite-set filters model uncertain cardinality and association. PHD filtering gives a first-moment recursion [9]; labeled RFS/LMB models retain identity [1, 2], and multi-sensor variants support cooperation [3]. Under unknown cross-correlations, KLA/GCI forms a normalized geometric mean [10], with PHD and multi-Bernoulli RFS formulations [11, 12]. Consensus LMB and labeled-RFS methods address networked tracking and label consistency [13, 14, 15, 16]; minimum-information-loss and arithmetic-average LMB rules provide alternative density-level objectives [17, 18]. We propose no new density-level KLA rule; we audit an implemented presence-conditioned per-label variant.

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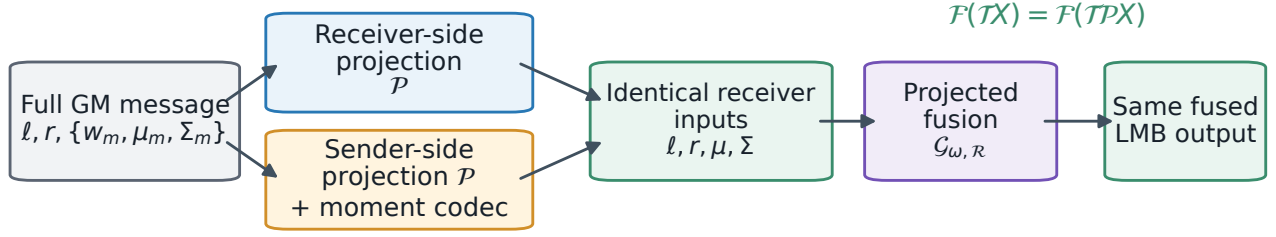


Fig. 1. Receiver-induced message factorization. The full path transports and decodes the GM before applying $\mathcal{P}$; the moment path applies $\mathcal{P}$ before value-preserving encoding of canonical binary64 fields. Under Proposition 1's contract, both paths present identical inputs to $\mathcal{G}_{\omega, \mathcal{R}}$ and produce the same fused LMB output.

Communication can be reduced at several layers. Event-triggered LMB methods control update transmission [4, 5, 6]; related triggers act on $M\delta$ -GLMB labels or hypotheses [19].

Partial-consensus PHD shares selected components [7]; state/covariance projection sends state, covariance, and time for asynchronous fusion [8]. Our tuple retains label/existence and certifies Gaussian overlap, missing-label normalization, and the codec.

3. RECEIVER-INDUCED MOMENT EXCHANGE

3.1. Canonical Label-Wise Projection

At time k , source j maintains an LMB posterior

$$\begin{aligned} \pi_j &= \{(r_j^\ell, p_j^\ell) : \ell \in \mathcal{L}_j\}, \\ p_j^\ell(x) &= \sum_{m=1}^{M_j^\ell} \tilde{w}_{jm}^\ell \mathcal{N}(x; m_{jm}^\ell, P_{jm}^\ell). \end{aligned} \quad (1)$$

The implementation maps nonfinite or negative component weights to zero and normalizes the remainder; it uses uniform weights only when all mapped weights vanish. Let $\mathcal{C}(A) = (A + A^\top)/2$. The canonical label-wise projection $\mathcal{P}$ retains (ℓ, r_j^ℓ) and computes

$$\mu_j^\ell = \sum_m \tilde{w}_{jm}^\ell m_{jm}^\ell, \quad (2)$$

Let $e_{jm}^\ell = m_{jm}^\ell - \mu_j^\ell$. Then

$$\Sigma_j^\ell = \mathcal{C} \left(\sum_m \tilde{w}_{jm}^\ell [\mathcal{C}(P_{jm}^\ell) + e_{jm}^\ell (e_{jm}^\ell)^\top] \right). \quad (3)$$

Thus, $\mathcal{P}$ replaces each mixture by the Gaussian with the same first two canonical moments. It is idempotent: $\mathcal{P}(\mathcal{P}(\pi)) = \mathcal{P}(\pi)$.

3.2. The Executed Presence-Conditioned Receiver

Receiver i aligns the union of source labels and applies $\mathcal{P}$ to each delivered source. Let $\mathcal{S}_i^\ell$ contain self and delivered neighbors carrying label ℓ . The confirmatory path uses base Metropolis weights γ_{ij} for both spatial and Bernoulli fusion, and applies KLA over the participating source subset for each active label:

$$\omega_{ij}^\ell = \frac{\gamma_{ij}}{\sum_{q \in \mathcal{S}_i^\ell} \gamma_{iq}}, \quad j \in \mathcal{S}_i^\ell. \quad (4)$$

This source-indexed, presence-conditioned rule is not a full-source LMB-density KLA that treats a missing label as $r^\ell = 0$. The broader software accepts distinct spatial and existence weights, but that generalized receiver is outside the KLA claim and frozen experiment.

The deterministic Cholesky regularizer $\mathcal{R}$ returns $\mathcal{C}(A)$ if factorization succeeds; otherwise it returns the first successful $\mathcal{C}(A) + 10^t \delta I$, where $\delta = 10^{-12} \max(1, \max |\text{diag}(A)|)$ and $t = 0, \dots, 15$. With $S_j^\ell = \mathcal{R}(\Sigma_j^\ell)$, the executed canonical update is

$$\begin{aligned} K_i^\ell &= \mathcal{R} \left(\sum_{j \in \mathcal{S}_i^\ell} \omega_{ij}^\ell (S_j^\ell)^{-1} \right), \\ h_i^\ell &= \sum_{j \in \mathcal{S}_i^\ell} \omega_{ij}^\ell (S_j^\ell)^{-1} \mu_j^\ell, \\ \bar{\mu}_i^\ell &= (K_i^\ell)^{-1} h_i^\ell, \quad \bar{\Sigma}_i^\ell = \mathcal{R}((K_i^\ell)^{-1}). \end{aligned} \quad (5)$$

Define $\lambda(A) = d \log(2\pi) + \log |A|$, evaluated from the Cholesky factor. The implementation's overlap factor is

$$\begin{aligned} \log \eta_{i, \mathcal{R}}^\ell &= -\frac{1}{2} \sum_{j \in \mathcal{S}_i^\ell} \omega_{ij}^\ell [(\mu_j^\ell)^\top (S_j^\ell)^{-1} \mu_j^\ell + \lambda(S_j^\ell)] \\ &\quad + \frac{1}{2} [(\bar{\mu}_i^\ell)^\top K_i^\ell \bar{\mu}_i^\ell + \lambda(\bar{\Sigma}_i^\ell)]. \end{aligned} \quad (6)$$

Without added jitter, Eq. (6) is the weighted Gaussian-product integral; the stated form is exact for the executed regularized

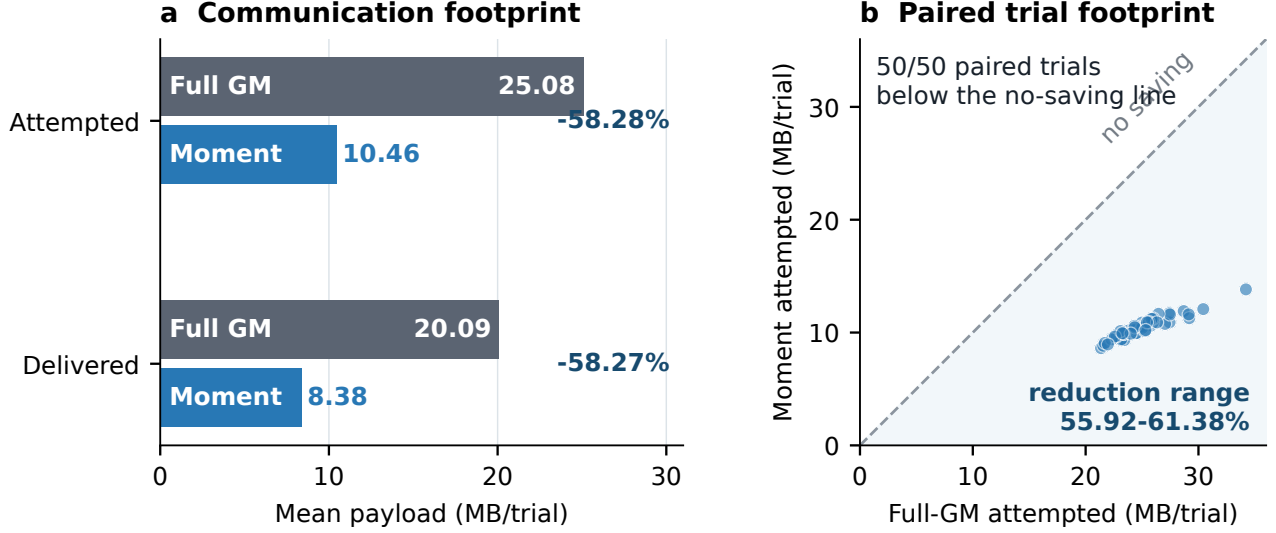


Fig. 2. Confirmatory communication evidence (50 paired trials). Left: mean attempted/delivered application-layer payloads; percentages are mean per-trial reductions. Right: each circle is one paired trial, and the dashed identity denotes no saving. Every trial lies below the identity, with attempted-byte reductions from 55.92% to 61.38%. Exact receiver-output equivalence is reported in Sec. 4.

path. With products below over $j \in \mathcal{S}_i^\ell$, the Bernoulli update is

$$q_1^\ell = \eta_{i,\mathcal{R}}^\ell \prod_j (r_j^\ell)^{\omega_{ij}^\ell}, \quad q_0^\ell = \prod_j (1 - r_j^\ell)^{\omega_{ij}^\ell}, \quad \bar{r}_i^\ell = \frac{q_1^\ell}{q_0^\ell + q_1^\ell}. \quad (7)$$

Each r_j^ℓ is clamped to $[10^{-9}, 1 - 10^{-9}]$, and the final denominator is lower-bounded by machine epsilon. Denote Eqs. (5)–(7) by $\mathcal{G}_{\omega,\mathcal{R}}$ and the receiver by $\mathcal{F}_{\omega,\mathcal{R}} = \mathcal{G}_{\omega,\mathcal{R}} \circ \mathcal{P}$.

Let $\mathcal{T}$ be the posterior-content map of the versioned encode/decode transport below. It preserves labels, r , weights, and means, and replaces each component covariance by $\mathcal{C}(P)$. Thus, on admissible messages, $\mathcal{P} \circ \mathcal{T} = \mathcal{P}$ and $\mathcal{T} \circ \mathcal{P} = \mathcal{P}$.

Proposition 1 (executed fusion-output equivalence). Assume identical active-label presence, base Metropolis weights, and delivery masks; common $\mathcal{C}$, $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{T}$; and disabled covariance inflation. Then

$$\mathcal{F}_{\omega,\mathcal{R}}(\mathcal{T}\pi_1, \dots, \mathcal{T}\pi_n) = \mathcal{F}_{\omega,\mathcal{R}}(\mathcal{T}\mathcal{P}\pi_1, \dots, \mathcal{T}\mathcal{P}\pi_n). \quad (8)$$

Proof. The identities above give $\mathcal{P}\mathcal{T}\pi_j = \mathcal{P}\mathcal{T}\mathcal{P}\pi_j = \mathcal{P}\pi_j$. They preserve each source-label’s r, μ, Σ and presence in $\mathcal{S}_i^\ell$. Hence the active ω_{ij}^ℓ , regularized S_j^ℓ , canonical K_i^ℓ, h_i^ℓ , overlap $\eta_{i,\mathcal{R}}^\ell$, and q_0^ℓ, q_1^ℓ are unchanged, yielding identical $\bar{r}_i^\ell, \bar{\mu}_i^\ell, \bar{\Sigma}_i^\ell$. $\square$

We call $\mathcal{P}(\pi)$ *fusion-sufficient* only for this receiver: members of its equivalence class produce the same output. Equation (8) does not imply equal posterior densities. It can fail for a mixture-aware consumer, different label selection or numerical maps, quantization, covariance inflation, or multi-round reuse.

3.3. Versioned Message and Byte Accounting

Both arms use the same versioned, little-endian binary codec underlying $\mathcal{T}$: a 24-byte message header, a 20-byte label header, and binary64 component fields for weight, mean, and the upper covariance triangle. The encoder applies $\mathcal{C}$ before packing, and the decoder reconstructs the symmetric binary64 covariance. For state dimension d , a message with M_ℓ components per label has

$$B = 24 + \sum_\ell \left[20 + 8M_\ell \left(1 + d + \frac{d(d+1)}{2} \right) \right] \text{ bytes.} \quad (9)$$

The full message retains every M_ℓ in Eq. (1); the proposed message sets $M_\ell = 1$ using Eqs. (2)–(3). Attempted bytes are counted after encoding and before the drop draw; delivered bytes count messages decoded after successful delivery. No tracker variable, scheduling decision, or fusion parameter changes.

4. EXPERIMENTS

4.1. Paired Confirmatory Protocol

We evaluate an eight-sensor 4+4 formation for 100 filtering steps per trial. Each node performs the same local LMB recursion and one synchronous neighbor- fusion round per step. The fixed graph has 16 undirected edges: two four-node cliques and four cross-clique links. Detection probability is 0.9, mean clutter is 3, and measurement-noise standard deviation is 3. Per-node drop probabilities $[0, 0.1, 0.2, 0.5]$,

with multiplicities $[1, 4, 1, 2]$, are randomly assigned each trial (mean 0.2).

We compare only periodic full GM fusion messages and fusion-sufficient moment messages. The 50 paired trials use seeds 82–131. Within each pair, target trajectories, measurements, schedules, labels, Metropolis weights, and drop uniforms are identical; attempted and delivered masks are checked for equality. Local accuracy uses Euclidean OSPA (cutoff $c = 5$, order $p = 2$), while agreement uses mean pairwise OSPA, position disagreement, and cardinality dispersion [20]. The primary statistic is the mean of 50 per-trial attempted-byte reductions. Its paired percentile-bootstrap interval uses 10,000 resamples and fixed seed 20270710.

Pre-confirmatory property tests independently recompute mixture moments and exercise asymmetric, ill-conditioned, and singular covariances. Codec tests round-trip canonical fields, while mutation tests reject perturbed labels, r , μ , or Σ . A negative control confirms that enabling covariance inflation breaks the claimed equivalence.

Table 1. Confirmatory means ($N = 50$). Att./Del. MB: attempted/delivered application payload per trial; Local/Cons.: OSPA to truth/mean pairwise sensor OSPA ($c = 5, p = 2$); lower is better.

Message	Att. MB	Del. MB	Local	Cons.
Full GM	25.084	20.091	2.076	1.951
Moment msg.	10.458	8.378	2.076	1.951

4.2. Results and Scope

Table 1 and Fig. 2 show a 58.277% mean per-trial reduction in attempted bytes (95% interval: 57.923–58.636%; minimum 55.922%). The mean delivered-byte reduction is 58.267%.

The field audit is stronger than rounded metric agreement. All 40,000 paired sensor-time snapshots have identical label sets. Across 1,119,037 matched label-instance comparisons, the maximum residuals in existence probability, mean, and covariance are each exactly zero. Consequently, local E-OSPA, consensus OSPA, position disagreement (4.614), and cardinality dispersion (0.203) are identical in every paired trial. This matches Proposition 1 and checks the serialization boundary.

5. DISCUSSION

Proposition 1 is simple only after certifying the receiver call chain. The full arm transports every GM component before applying $\mathcal{P}$; the moment arm projects before transport. The identities $\mathcal{PT} = \mathcal{P}$ and $\mathcal{TP} = \mathcal{P}$ include the codec’s covariance canonicalization. Both arms then execute identical label alignment, $\mathcal{R}$, and Bernoulli fusion. This crosses the serialization boundary rather than projecting both arms in advance. Independent moment oracles, field mutations, and a

covariance-inflation negative control verify that the audit detects contract violations.

Equation (9) explains why 58.28% is workload specific. Let $c_d = 1 + d + d(d + 1)/2$. For fixed labels, the saved payload is $8 \sum_{\ell} (M_{\ell} - 1) c_d$ bytes; headers remain. Savings vanish at $M_{\ell} = 1$ and grow with dimension and mixture multiplicity. The header fraction limits small messages, so the interval describes this workload rather than a universal compression ratio.

The equivalence fails under gating that uses modality, weights derived from mixtures, different label pruning, quantization, covariance inflation, or later reuse of individual modes. Independent implementations may also differ in summation or regularization. Exactness therefore requires common $\mathcal{C}$, $\mathcal{P}$, $\mathcal{R}$, $\mathcal{T}$, and canonical binary64 fields. The broader software’s distinct spatial and existence weights define a generalized update outside this KLA claim.

The byte boundary is the encoded application payload; it excludes framing, headers, fragmentation, retransmission, latency, and radio energy. Payload-size-independent loss keeps delivery masks matched. Link-level benefits require measurements on the target packetization and hardware.

This case suggests a receiver-driven procedure: identify the consumer’s first irreversible map, certify that the remaining call chain factors through it, promote its output to a message type, and test after serialization. Unlike scheduling, component selection, or generic state/covariance projection, our certificate covers one consumer’s labeled Bernoulli output, including existence overlap and missing-label semantics. Evidence is limited to one homogeneous eight-sensor model, graph, binary64 codec, and synchronous round. Other dimensions, mixtures, sensors, links, packetizations, and multi-round consumers remain future tests.

6. CONCLUSION

We derived a receiver-induced moment message from an executed projected KLA-LMB receiver. Sender-side projection preserved every audited fusion output and reduced attempted application-layer bytes by 58.28% across 50 paired trials. The certificate includes label presence, Metropolis weights, numerical regularization, and canonical transport. It is exact for this receiver, not a general equivalence result for mixture-aware or multi-round KLA fusion.

COMPLIANCE WITH ETHICAL STANDARDS

This numerical simulation study required no ethical approval.

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